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Assignment: Covariance & Correlation

Assignment

Q1. Define Covariance and explain how it differs from Correlation in terms of scale and interpretation.

Sol)

- Covariance only tells direction not strength.
- It tells us the direction of relationship (Positive & Negative).

Difference Between Covariance and Correlation:

Basis	Covariance	Correlation
Scale	Depends on units	Unit-free
Range	No fixed range	-1 to +1
Interpretation	Hard to interpret magnitude	Easy to interpret strength

Q2. What does a positive, negative, and zero covariance indicate about the relationship between two variables?

Sol)

***Positive Covariance* →** Variables move in same direction
(If X increases, Y increases)

***Negative Covariance* →** Variables move in opposite direction
(If X increases, Y decreases)

***Zero Covariance* →** No linear relationship.

Q3. Discuss the limitations of covariance as a measure of relationship between two variables. Why is correlation preferred in many cases?

Sol)

Limitations:

1. No fixed range
2. Depends on measurement units
3. Difficult to interpret strength
4. Cannot compare different datasets

That's why correlation is preferred because it is standardized and ranges from -1 to +1.

Q4. Explain the difference between Pearson's correlation coefficient and Spearman's rank correlation coefficient. When would you prefer to use Spearman's correlation?

Sol)

Pearson

- Measures linear relationship
- Uses actual values
- Sensitive to outliers
- Requires normal distribution

Spearman

- Measures monotonic relationship
- Uses ranks
- Less sensitive to outliers
- No strict distribution assumption

When to Use Spearman:

- When data is not normally distributed
- When dealing with ranks
- When relationship is non-linear but monotonic

Q5. If the correlation coefficient between two variables X and Y is 0.85, interpret this value in context. Can you infer causation from this value? Why or why not?

Sol)

0.85 indicates ***strong positive correlation.***

As X increases, Y increases strongly.

We cannot infer causation from this value

Because:

- Correlation does not imply causation
- There may be third variable involved